

CHAPTER 14

Breathing and Exchange of Gases

Human Respiratory System

- Select the sequence of steps in Respiration. (2023-Manipur)
 - Diffusion of gases (O_2 and CO_2) across alveolar membrane.
 - Diffusion of O_2 and CO_2 between blood and tissues.
 - Transport of gases by the blood.
 - Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out.
 - Utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2 .

Choose the **correct** answer from the options given below:

- (D), (A), (C), (B), (E)
- (C), (B), (A), (E), (D)
- (B), (C), (E), (D), (A)
- (A), (C), (B), (E), (D)

Mechanism of Breathing

- Select the correct events that occur during inspiration (2020)
 - Contraction of diaphragm
 - Contraction of external inter-costal muscles
 - Pulmonary volume decreases
 - Intra pulmonary pressure increases
 - (3) and (4)
 - (1), (2) and (4)
 - Only (4)
 - (1) and (2)
- Lungs do not collapse between breaths and some air always remains in the lungs which can never be expelled because: (2016 - II)
 - There is a positive intrapleural pressure
 - Pressure in the lungs is higher than the atmospheric pressure.
 - There is a negative pressure in the lungs.
 - There is a negative intrapleural pressure pulling at the lung walls

Respiratory Volumes & Capacities

- Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Residual Volume	(I)	Maximum volume of air that can be breathed in after forced expiration
(B)	Vital Capacity	(II)	Volume of air inspired or expired during normal respiration

(C)	Expiratory Capacity	(III)	Volume of air remaining in lungs after forcible expiration
(D)	Tidal Volume	(IV)	Total volume of air expired after normal inspiration

Choose the **correct** answer from the options given below:

- A-IV, B-III, C-II, D-I
- A-II, B-IV, C-I, D-III
- A-III, B-I, C-IV, D-II
- A-I, B-II, C-III, D-IV

- Match List-I with List-II: (2024)

List-I		List-II	
(A)	Expiratory capacity	(I)	Expiratory reserve volume + Tidal volume + Inspiratory reserve volume
(B)	Functional residual capacity	(II)	Tidal volume + Expiratory reserve volume
(C)	Vital capacity	(III)	Tidal volume + Inspiratory reserve volume
(D)	Inspiratory capacity	(IV)	Expiratory reserve volume + Residual volume

Choose the correct answer from the options given below :

- A-(II), B-(I), C-(IV), D-(III)
 - A-(I), B-(III), C-(II), D-(IV)
 - A-(II), B-(IV), C-(I), D-(III)
 - A-(III), B-(II), C-(IV), D-(I)
- Vital capacity of lung is _____. (2023)
 - IRV + ERV + TV
 - IRV + ERV
 - IRV + ERV + TV + RV
 - IRV + ERV + TV - RV
 - Which of the following statements are correct with respect to vital capacity? (2022 Re)
 - It includes ERV, TV and IRV.
 - Total volume of air a person can inspire after a normal expiration.
 - The maximum volume of air a person can breathe in after forced expiration.
 - It includes ERV, RV and IRV.
 - The maximum volume of air a person can breathe out after a forced inspiration.

Choose the most appropriate answer from the options given below.

- (A) and (E)
- (B), (D) and (E)
- (A), (C) and (D)
- (A), (C) and (E)

8. The Total Lung Capacity (TLC) is the total volume of air accommodated in the lungs at the end of a forced inspiration. This includes: (2020-Covid)
- RV; ERV; IC and EC
 - RV; ERV; VC (Vital Capacity) and FRC (Functional Residual Capacity)
 - RV (Residual volume); ERV (Expiratory Reserve Volume); TV (Tidal Volume); and IRV (Inspiratory Reserve Volume)
 - RV; IC (Inspiratory Capacity); EC (Expiratory Capacity); and ERV
9. Tidal Volume and Expiratory Reserve Volume of an athlete is 500 mL and 1000 mL, respectively. What will be his Expiratory Capacity if the Residual Volume is 1200 mL? (2019)
- 1500 mL
 - 1700 mL
 - 2200 mL
 - 2700 mL
10. Match the items given Column-I with those in Column-II and select the correct option given below (2018)
- | | Column-I | | Column-II |
|----|----------------------------|------|--------------|
| A. | Tidal volume | i. | 2500-3000 mL |
| B. | Inspiratory Reserve volume | ii. | 1100-1200 mL |
| C. | Expiratory Reserve volume | iii. | 500-550 mL |
| D. | Residual volume | iv. | 1000-1100 mL |
- A-iii B-ii C-i D-iv
 - A-iii B-i C-iv D-ii
 - A-i B-iv C-ii D-iii
 - A-iv B-iii C-ii D-i
11. Lungs are made up of air-filled sacs the alveoli. They do not collapse even after forceful expiration, because of: (2017-Delhi)
- Residual Volume
 - Inspiratory Reserve Volume
 - Tidal Volume
 - Expiratory Reserve Volume
13. Which of the following factors are favourable for the formation of oxyhaemoglobin in alveoli? (2024)
- Low $p\text{CO}_2$ and High H^+ concentration
 - Low $p\text{CO}_2$ and High temperature
 - High $p\text{O}_2$ and High $p\text{CO}_2$
 - High $p\text{O}_2$ and Lesser H^+ concentration
14. Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli. (2021)
- Low $p\text{O}_2$, high $p\text{CO}_2$, more H^+ , higher temperature
 - High $p\text{O}_2$, high $p\text{CO}_2$, less H^+ , higher temperature
 - Low $p\text{O}_2$, low $p\text{CO}_2$, more H^+ , higher temperature
 - High $p\text{O}_2$, low $p\text{CO}_2$, less H^+ , lower temperature
15. The partial pressures (in mm Hg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are: (2021)
- $p\text{O}_2 = 40$ and $p\text{CO}_2 = 45$
 - $p\text{O}_2 = 95$ and $p\text{CO}_2 = 40$
 - $p\text{O}_2 = 159$ and $p\text{CO}_2 = 0.3$
 - $p\text{O}_2 = 104$ and $p\text{CO}_2 = 40$
16. Identify the wrong statement with reference to transport of oxygen. (2020)
- Partial pressure of CO_2 can interfere with O_2 binding with haemoglobin.
 - Higher H^+ conc. in alveoli favours the formation of oxyhaemoglobin.
 - Low $p\text{CO}_2$ in alveoli favours the formation of oxyhaemoglobin.
 - Binding of oxygen with haemoglobin is mainly related to partial pressure of O_2 .
17. The partial pressure of oxygen in the alveoli of the lungs is: (2016 - II)
- Less than that in the blood
 - Less than that of carbon dioxide
 - Equal to that in the blood
 - More than that in the blood
18. Reduction in pH of blood will: (2016 - I)
- Reduce the rate of heart beat
 - Reduce the blood supply to the brain
 - Decrease the affinity of hemoglobin with oxygen
 - Release bicarbonate ions by the liver
19. When you hold your breath, which of the following gas changes in blood would first lead to the urge to breathe? (2015 Re)
- Rising CO_2 concentration
 - Rising CO_2 and falling O_2 concentration
 - Falling O_2 concentration
 - Falling CO_2 concentration
20. Approximately seventy percent of carbon-dioxide absorbed by the blood will be transported to the lungs: (2014)
- As carbamino-haemoglobin
 - As bicarbonate ions
 - In the form of dissolved gas molecules
 - By binding to R.B.C.

Exchange and Transport of Gases

12. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**. (2024 Re)
- Assertion (A):** During the transportation of gases, about 20-25 percent of CO_2 is carried by Haemoglobin as carbamino-haemoglobin.
- Reason (R):** This binding is related to high $p\text{CO}_2$ and low $p\text{O}_2$ in tissues.
- In the light of the above statements, choose the correct answer from the option given below:
- A is true but R is false.
 - A is false but R is true.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true but R is NOT the correct explanation of A.

Regulation & Disorders of Respiratory System

21. Match the following columns and select the correct option:
(2020-Covid)

	Column-I		Column-II
1.	Pneumotaxic Centre	(i)	Alveoli
2.	O ₂ Dissociation curve	(ii)	Pons region of brain
3.	Carbonic Anhydrase	(iii)	Haemoglobin
4.	Primary site of exchange of gases	(iv)	R.B.C.

- (1) (2) (3) (4)
- a. (ii) (iii) (iv) (i)
- b. (iii) (ii) (iv) (i)
- c. (iv) (i) (iii) (ii)
- d. (i) (iii) (ii) (iv)

22. Due to increasing air-borne allergens and pollutants, many people in urban areas are suffering from respiratory disorder causing wheezing due to (2019)
- Benign growth on mucous lining of nasal cavity
 - Inflammation of bronchi and bronchioles
 - Proliferation of fibrous tissues and damage of the alveolar walls
 - Reduction in the secretion of surfactants by pneumocytes.

23. Which of the following options correctly represents the lung conditions in asthma and emphysema, respectively? (2018)
- Inflammation of bronchioles; Decreased respiratory surface
 - Increased number of bronchioles; Increased respiratory surface
 - Increased respiratory surface; Inflammation of bronchioles
 - Decreased respiratory surface; Inflammation of bronchioles
24. Which of the following is an occupational respiratory disorder? (2018)
- Anthraxis
 - Silicosis
 - Botulism
 - Emphysema
25. Asthma may be attributed to: (2016 - I)
- Bacterial infection of the lungs
 - Allergic reaction of the mast cells in the lungs
 - Inflammation of the trachea
 - Accumulation of fluid in the lungs
26. Name the chronic respiratory disorder caused mainly by cigarette smoking: (2016 - I)
- Emphysema
 - Asthma
 - Respiratory acidosis
 - Respiratory alkalosis
27. Name the pulmonary disease in which alveolar surface area involved in gas exchange is drastically reduced due to damage in the alveolar walls. (2015 Re)
- Emphysema
 - Pneumonia
 - Asthma
 - Pleurisy

Answer Key

1. (a) 2. (d) 3. (d) 4. (c) 5. (c) 6. (a) 7. (d) 8. (c) 9. (a) 10. (b)
11. (a) 12. (c) 13. (d) 14. (d) 15. (d) 16. (b) 17. (d) 18. (c) 19. (a) 20. (b)
21. (a) 22. (b) 23. (a) 24. (b) 25. (b) 26. (a) 27. (a)